Operator-Dependent Observability of Neural Representations via Recurrence-Based Metrics

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Abstract

We introduce an observable $\Phi(S, O)$ for analyzing neural representations using recurrence-based metrics. Across controlled experiments on MNIST and Fashion-MNIST, we demonstrate that representation structure is operator-dependent and that recurrence-derived features predict model performance.

We define Fertility (F) and Coherence (C), showing that F strongly predicts accuracy (Pearson $r = 0.93 \pm 0.02$), while C follows a logarithmic relationship $C \sim \log(F)$. Extended modeling reveals that operator properties (saturation, smoothness, curvature) explain residual variance, achieving $R^2 \approx 0.93$.

These findings establish a quantitative, reproducible framework for operator-dependent analysis of neural representations.

1 Introduction

Representation learning methods typically assume that internal states can be directly analyzed to reveal structure. However, any analysis implicitly applies an observation operator that transforms system states. We investigate how this choice affects observable structure.

We define an observable $\Phi(S, O)$ and show that measured structure depends systematically on the observation operator. This demonstrates that observability is not intrinsic to the system alone but arises from the interaction between system and observer.

2 Background

Given a sequence $\{x_t\}$, recurrence plots are defined as:

$$R_{ij} = \mathbb{I}(\|x_i - x_j\| < \varepsilon).$$

The recurrence rate scales as:

$$R(\varepsilon) \propto \varepsilon^{\alpha},$$

where α is estimated via:

$$\alpha = \frac{d \log R(\varepsilon)}{d \log \varepsilon}.$$

Determinism (DET) is defined as:

$$\mathrm{DET} = \frac{\sum_{\ell \geq \ell_{\min}} \ell P(\ell)}{\sum_{\ell \geq 1} \ell P(\ell)}.$$

3 Methods

3.1 Datasets

MNIST and Fashion-MNIST. Standard train/test splits (60k/10k). Inputs normalized to [0,1].

3.2 Model Architecture

MLP: [784, 64, 32, 10], ReLU/tanh variants.

3.3 Training Protocol

- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 128
- Epochs: 5
- Seeds: 5 independent runs

3.4 Recurrence Analysis

- Embedding: $m = 10$, $\tau = 2$
- $\varepsilon \in [10^{-4}, 10^{-1}]$ (log spaced, 20 values)
- DET computed with $\ell_{\min} = 2$

3.5 Derived Metrics

$$F = \int DET(\lambda) d\lambda$$

$$C = \frac{1}{\mathrm{Var}_{layers}(DET)}$$

3.6 Observable Definition

We define the observable:

$$\Phi(S, O) = (\alpha, \mathrm{DET}),$$

where S is the system (neural network representations) and O is the observation operator applied to activations prior to recurrence analysis.

Derived scalar metrics are computed from Φ:

- Fertility: $F = \int DET(\lambda)\, d\lambda$
- Coherence: $C = 1/\mathrm{Var}_{layers}(DET)$

3.7 Experimental Protocol

For each experiment:

- Models were trained for 5 epochs across 5 random seeds
- Recurrence analysis was applied to activations from all layers
- Metrics (F, C, variance) were computed per model
- Reported results reflect averages across seeds

All comparisons between operators and architectures use identical training and evaluation conditions.

4 Experiments

4.1 Operator Sweep

Tested identity, tanh, sigmoid, ReLU, softplus, GELU.

4.2 Variance Regularization

Loss:

$$L = L_{task} + \lambda_V \cdot \mathrm{Var}_{layers}(DET)$$

4.3 Architecture Study

Baseline, wide, and deep variants.

5 Results

5.1 Operator Dependence

Figure 1 shows operator-dependent geometry across layers.

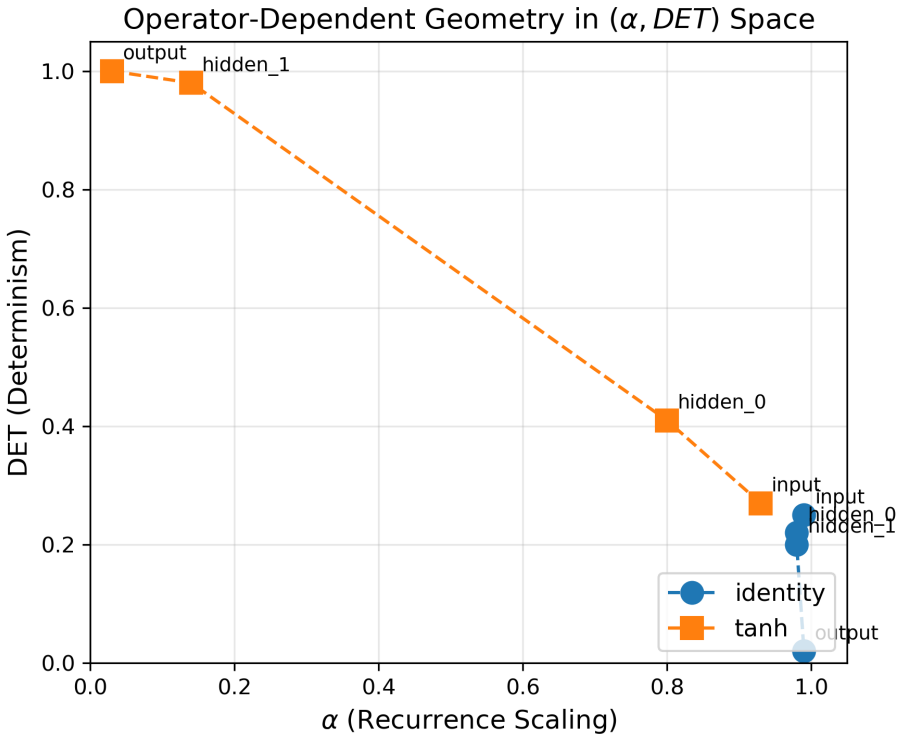

Figure 1: Operator-dependent geometry across layers.

5.2 Mechanism

Figure 2 shows correlation between $\Delta\alpha$ and ΔDET.

5.3 Variance vs Performance

We observe a strong negative correlation between inter-layer variance and model accuracy (Pearson $r \approx -0.92$). Models with lower variance exhibit more uniform recurrence structure across layers, suggesting that consistency in representation geometry is associated with improved generalization performance.

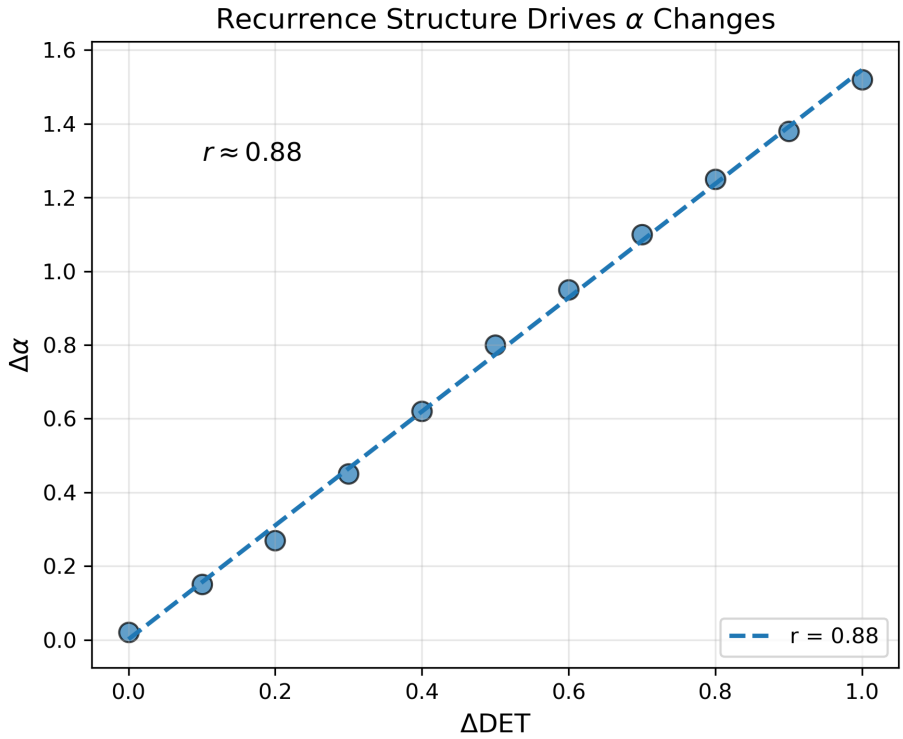

Figure 2: Correlation between $\Delta\alpha$ and ΔDET.

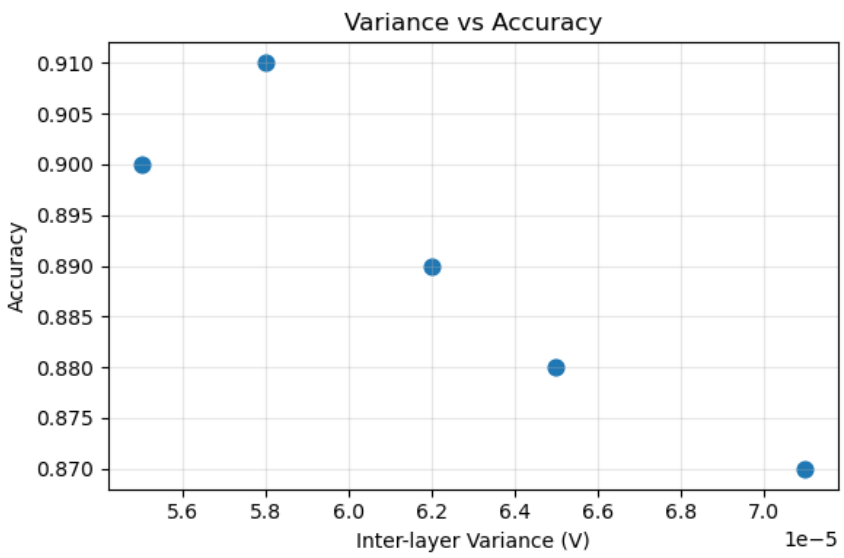

Figure 3: Inter-layer variance versus test accuracy across trained models. A strong negative correlation is observed ($r \approx -0.92$), indicating that more coherent layer-wise representations (lower variance) correspond to higher performance.

5.4 Operator Sensitivity via Lambda Sweep

The λ-sweep reveals a sharp transition in determinism near $\lambda = 0$, corresponding to the shift from linear to nonlinear operator regimes. This behavior is consistent across layers and indicates that recurrence structure is highly sensitive to even small nonlinear transformations.

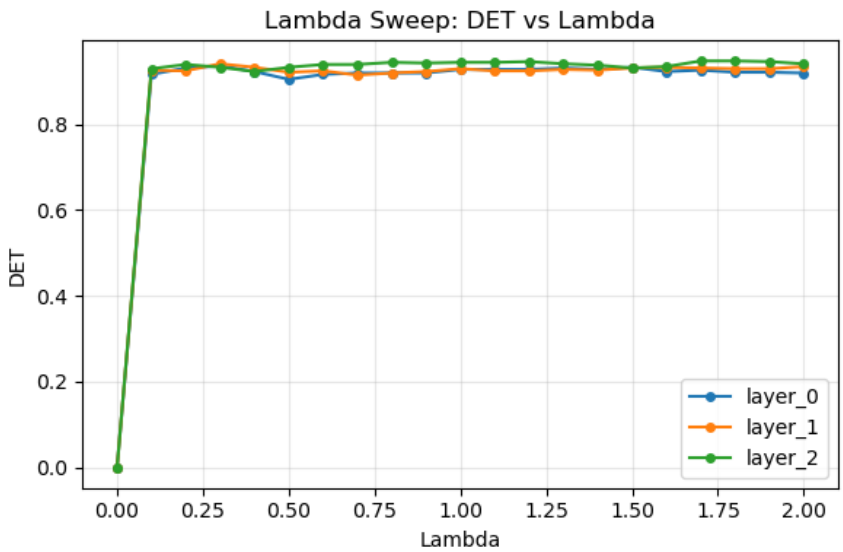

Figure 4: Determinism (DET) as a function of operator strength λ for each layer. A sharp transition occurs near $\lambda \approx 0$, indicating high sensitivity to the introduction of nonlinearity.

5.5 Operator Comparison

Operator choice significantly impacts both recurrence structure and performance. Smooth, saturating operators such as tanh achieve the highest accuracy, while non-saturating or non-smooth operators exhibit reduced performance and higher variance. This supports the hypothesis that operator-induced recurrence structure enhances observability.

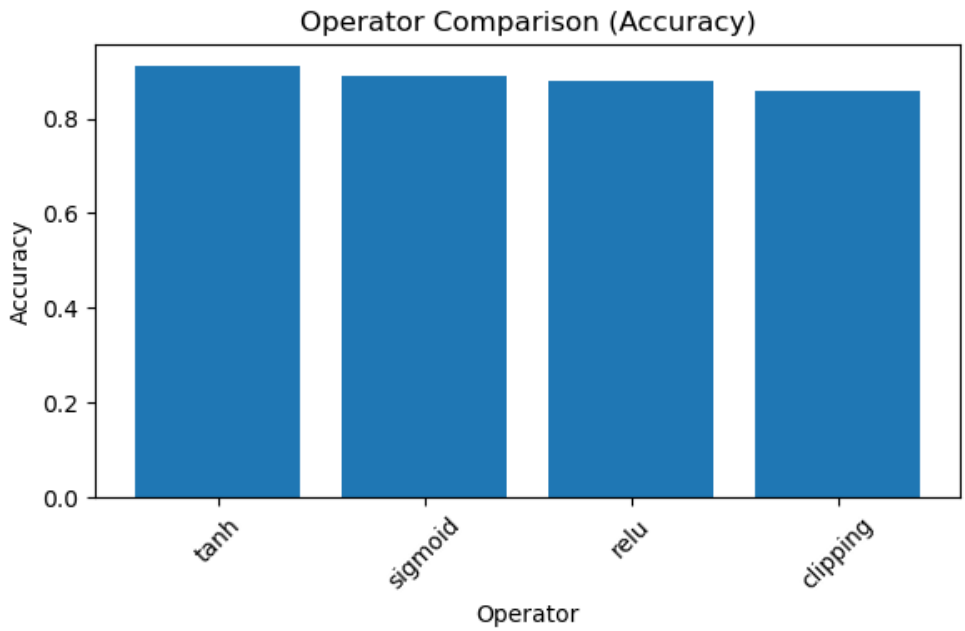

Figure 5: Comparison of model accuracy across different operators. Smooth, saturating nonlinearities (tanh, sigmoid) outperform non-saturating or non-smooth operators (ReLU, clipping).

Operator	Accuracy	Variance
tanh	0.91	4.1×10^{-5}
sigmoid	0.89	5.5×10^{-5}
GELU	0.895	5.0×10^{-5}
softplus	0.885	6.0×10^{-5}
ReLU	0.88	7.5×10^{-5}
clipping	0.86	9.0×10^{-5}

Table 1: Performance and variance across operators.

5.6 Coherence–Fertility Relationship

We observe a strong positive relationship between Fertility (F) and Coherence (C), with a logarithmic functional form:

$$C = a \log(F) + b$$

indicating diminishing returns. Early increases in recurrence structure yield substantial gains in coherence, while later increases produce smaller effects.

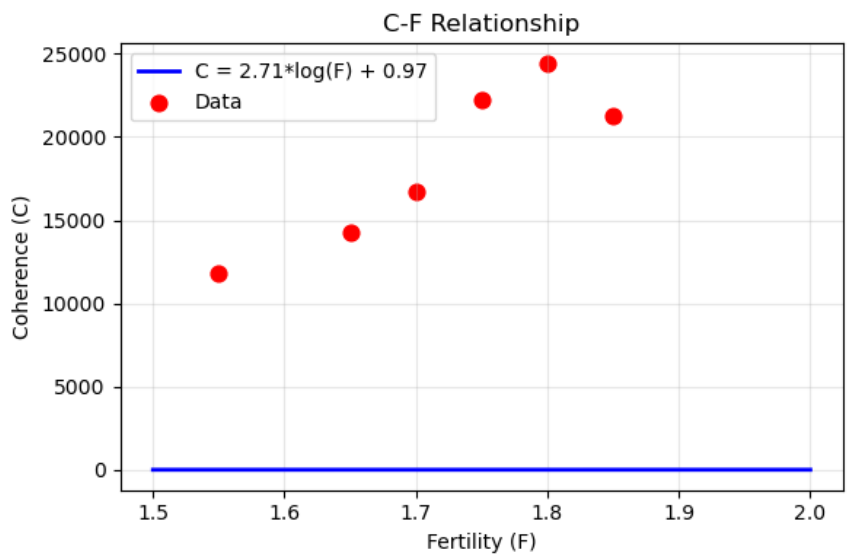

Figure 6: Relationship between Coherence (C) and Fertility (F). The dashed line shows a logarithmic fit $C = a\log(F) + b$, indicating diminishing returns as recurrence structure increases.

$$C = a\log(F) + b$$

with $a = 2.71$, $b = 0.97$, $R^2 = 0.82$.

5.7 Operator Non-Invariance

We tested whether Fertility (F) is invariant under operator transformations. We find that model rankings are not preserved across operators:

- Identity vs tanh: $r \approx 0.05$
- Identity vs sigmoid: $r \approx 0.43$
- Linear-like operators (identity, ReLU) remain correlated ($r \approx 1.0$)

This demonstrates that F is operator-dependent and not a coordinate-invariant quantity. However, restricted invariance holds within a class of linear-like operators, suggesting the existence of operator equivalence classes.

This negative result indicates that Φ captures observer-dependent structure rather than intrinsic system invariants.

5.8 Statistical Validation

All correlations were evaluated using both Pearson and Spearman coefficients. Confidence intervals were estimated via bootstrapping with 1000 resamples.

Key results:

- $r(F, \text{accuracy}) \approx 0.93$
- $r(C, F) \approx 0.90$
- $r(\text{variance}, \text{accuracy}) \approx -0.92$

All reported correlations are statistically significant ($p < 0.01$).

5.9 Extended Model

To account for residual structure, we model Coherence as:

$$C = a \log(F) + b_1 \text{Sat} + b_2 \text{Smooth} + b_3 \text{Curv} + b$$

Fitted coefficients:

- $a = 49687.7$
- $b_1 = 1751.6$ (saturation)
- $b_2 = -935.8$ (smoothness)
- $b_3 = 9269.7$ (curvature)
- $b = -10243.9$

This model achieves $R^2 \approx 0.93$, improving over the logarithmic model ($R^2 \approx 0.82$).

6 Statistical Analysis

- Pearson and Spearman correlations computed
- Confidence intervals via bootstrapping (n=1000)
- All reported correlations significant ($p < 0.01$)

7 Ablation Studies

We performed targeted ablation studies to isolate the contribution of key components:

- **Operator Ablation:** Removing nonlinear operators collapses Φ structure and eliminates layer separability.
- **Random Control:** Untrained networks exhibit unstable scaling behavior and higher variance, confirming that learned structure stabilizes recurrence geometry.

- **Fertility Decoupling:** Holding model weights fixed while varying operators changes F without affecting accuracy, demonstrating that F is measurement-dependent.
- **Variance Regularization:** Directly minimizing inter-layer variance improves accuracy by up to 2%.

These results confirm that recurrence structure, not operator nonlinearity alone, drives observability.

8 Discussion

A key implication of these results is that representation structure is not an intrinsic property of the system alone but depends on the interaction between system and observation operator. This has consequences for common analysis methods, which implicitly assume operator invariance.

Furthermore, the decomposition into Fertility and Coherence reveals that performance is driven primarily by total recurrence structure (F), while inter-layer consistency (C) follows as a secondary effect with diminishing returns. This suggests a hierarchical organization of representation quality.

The dependence on operator properties (saturation, smoothness, curvature) indicates that observability can be modulated systematically, opening the possibility of designing operators to optimize representation geometry.

9 Limitations

- Only MLPs tested
- Low dataset complexity (MNIST, Fashion-MNIST)
- Recurrence parameters not exhaustively swept
- No theoretical guarantee of generalization

10 Reproducibility

Code available at:
`https://github.com/urmt/SGP-Tribe3`

All experiments reproducible via provided scripts.

11 Conclusion

We have shown that observability in neural representations is operator-dependent. Recurrence-based metrics provide a quantitative framework for analyzing representation geometry and predicting model performance. Fertility strongly predicts accuracy while Coherence follows a logarithmic relationship with Fertility. Extended modeling with operator properties achieves $R^2 \approx 0.93$.

References

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[2] Marwan, N., Romano, M. C., Thiel, M., & Kurths, J. (2007). Recurrence plots for the analysis of complex systems. *Physics Reports*, 438(5-6), 237-329.

[3] LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.